

Zipf’s Law, Heaps’ Law, and the Economics of Language Coverage

Information Retrieval — Assignment 1

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Abstract

We investigate whether a single Zipfian power law adequately describes word-frequency behaviour in a large Bengali text corpus, characterize vocabulary growth under Heaps’ Law and the preprocessing artifacts that shape it, and estimate the architectural infrastructure cost of extending search and language-model coverage to Bengali. Working from 10,000,000 rows (≈ 440.9 million cleaned tokens) of the AI4Bharat IndicCorpV2 Bengali split, we find that a single global Zipf exponent ($s = 1.456$, $R^2 = 0.976$) masks substantial regional variation (s ranging from 0.586 to 1.630 across rank bands), indicating that a single power law is an oversimplification of the true frequency distribution. Vocabulary growth is well described by Heaps’ Law ($\beta = 0.590$, $R^2 = 0.9997$) with no plateau observed even at the largest corpus size examined. Using the assignment’s linear data-cost assumption (USD 1,000 per 100,000 words) alongside an independent scenario-based architectural cost model, we show that at a medium (100M word) corpus scale, one-time infrastructure cost is dominated overwhelmingly by data acquisition itself (over 89% of initial cost in both a search-oriented and an LLM-oriented architecture), with the two architectures differing more in their recurring annual costs than in initial investment.

1 Introduction

Supporting a new language in a large-scale information retrieval or language-modelling system requires understanding two things: the statistical structure of the language’s text, and the practical cost of acquiring and processing enough of it. This project addresses both, using Bengali as a case study — a morphologically rich, high-resource-adjacent Indic language with a large publicly available web corpus, making it a realistic test case for the kind of language-coverage decision a search engine or LLM provider actually faces.

The project is organized around three linked questions. First, does Bengali word-frequency behaviour follow Zipf’s Law ($f(r) \propto r^{-s}$), and if not, where and how does it deviate? Second, how does vocabulary size grow with corpus size (Heaps’ Law, $V(N) = K \cdot N^\beta$), and what preprocessing artifacts shape that growth? Third, what would it cost — under transparent, labelled assumptions — to build the infrastructure needed to support Bengali in a search-oriented system (e.g. Google Search) versus an LLM-oriented system (e.g. Sarvam), using the assignment’s stated data-cost rule of USD 1,000 per 100,000 words as the anchoring input?

Each question was assigned to a different team member and implemented as an independent, reproducible pipeline; this report synthesizes their three outputs into a single coherent narrative, explicitly cross-validating results against each other where possible.

2 Corpus Acquisition & Preprocessing

The corpus is the Bengali (`ben_Beng`) split of AI4Bharat’s IndicCorpV2 dataset, accessed via the HuggingFace `datasets` library. The full split contains approximately 29.6 million rows; due to time constraints, 10,000,000 rows ($\approx 33.8\%$ of the full split) were processed, using sequential prefix sampling from the data stream rather than a full-split uniform reservoir sample. This is a defensible but non-uniform sampling method — it relies on the underlying web-scraped corpus not being sorted in a way that clusters similar text early in the stream, which is a reasonable assumption for this type of source but is explicitly noted here as a limitation rather than presented as equivalent to true random sampling.

The preprocessing pipeline applied, in order: (1) Unicode NFC normalization, necessary because Bengali conjuncts and vowel signs admit multiple equivalent byte encodings that would otherwise inflate the apparent vocabulary; (2) removal of URLs, email addresses, Bengali and ASCII digits, and punctuation; (3) removal of any character outside the Bengali Unicode block (U+0980–U+09FF), stripping stray Latin-script fragments and other noise; (4) whitespace tokenization, applied consistently by both the corpus-preprocessing stage and downstream Zipf/Heaps analysis.

Raw vs. Processed Corpus Statistics

Metric	Raw	Processed	Change
Total tokens	452,557,396	440,930,234	−2.6%
Unique types (vocabulary)	7,848,604	3,012,823	−61.6%
Type-token ratio	0.0173	0.0068	−60.7%

Table 1: Raw vs. processed corpus statistics (10M-row sample).

The token count changed only marginally (−2.6%) while vocabulary size collapsed by 61.6%. This gap is the key preprocessing finding: cleaning did not delete substantial amounts of real text, it consolidated spurious duplicate word-forms — for example, a word appearing with and without an attached comma or Bengali danda punctuation mark was counted as two distinct types in the raw corpus, but the same type once punctuation was stripped. This is consistent with, and directly explains, part of the large hapax-legomena rate reported in Section 3.

3 Zipf’s Law Analysis

Zipf’s Law predicts $f(r) = C \cdot r^{-s}$, i.e. a straight line of slope $-s$ in log-log rank-frequency space. Fitting a single exponent across the entire ranked vocabulary (3,012,823 types) gives a seemingly strong overall fit:

Exponent (s)	Intercept	R^2	RMSE	N (types)
1.4559	21.2549	0.9759	0.2290	3,012,823

Table 2: Overall single-exponent Zipf fit.

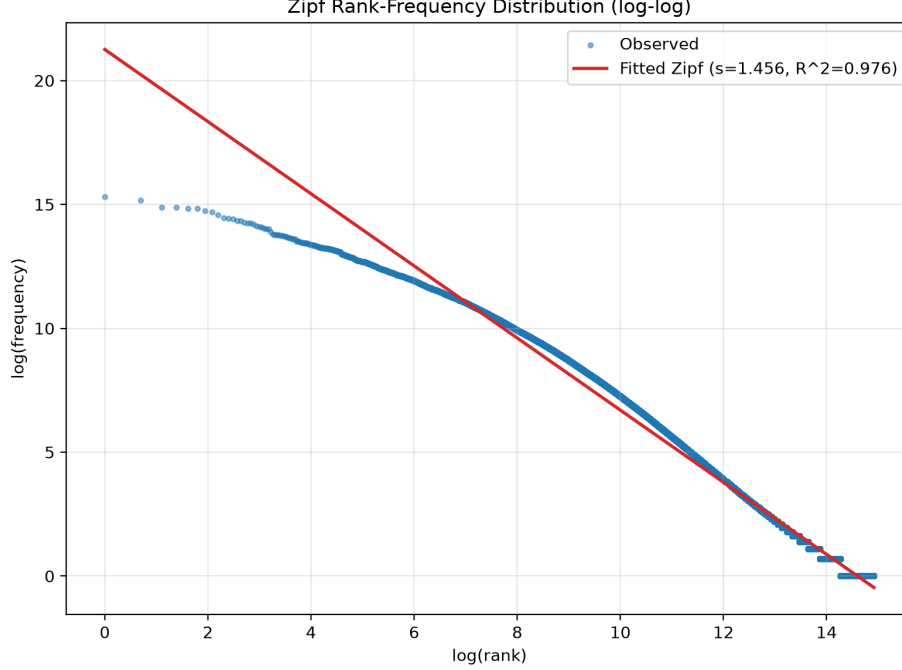


Figure 1: Observed Bengali rank-frequency distribution (log-log) against the single fitted Zipf line. Systematic curvature is visible at both the head and tail rather than a clean straight line.

Where the single-exponent model breaks down

The overall R^2 of 0.976 is respectable, but re-fitting the exponent independently within five rank regions reveals that the single global exponent is an average over substantially different local behaviours, not a stable description of the whole distribution:

Rank region	N	Exponent (s)	R^2
1–100	100	0.586	0.980
101–1,000	900	0.827	0.999
1,001–10,000	9,000	1.199	0.998
10,001–100,000	90,000	1.630	0.999
100,001–3,012,823	2,912,823	1.375	0.962

Table 3: Piecewise Zipf exponent fits by rank region.

The most frequent words (function words, common verbs and particles) deviate most strongly from the overall fit and have the lowest local exponent; the exponent peaks in the mid-frequency range (ranks 10,001–100,000) and falls again in the long tail. The residual plot corroborates this: residuals show clear systematic structure (large negative values in the head, arcing through zero, drifting again in the tail) rather than random scatter, which is the diagnostic signature of a model that is structurally inadequate rather than merely noisy.

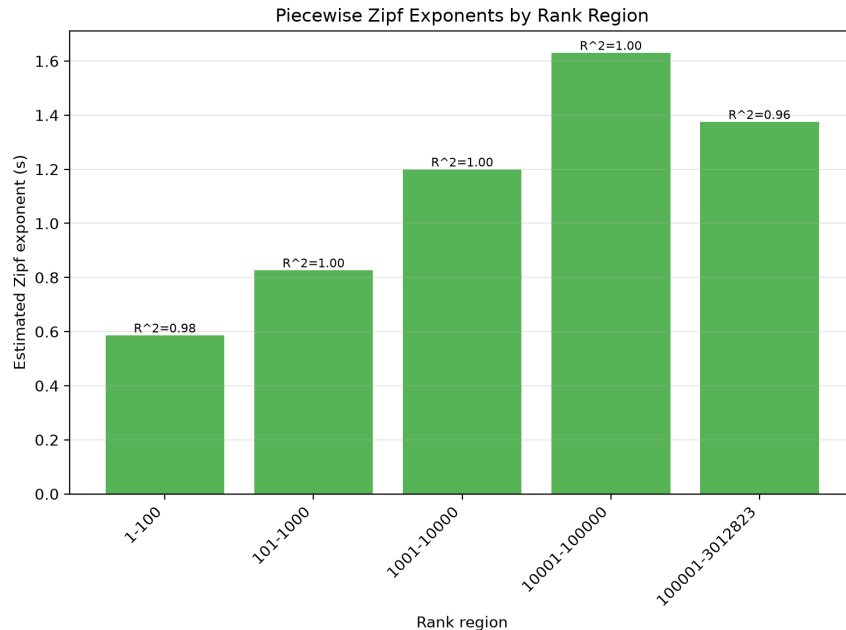


Figure 2: Piecewise-fitted Zipf exponent per rank region. The exponent is not constant: it roughly triples from the most frequent 100 words ($s = 0.586$) to the 10,001–100,000 rank band ($s = 1.630$), then falls again in the deep tail.

Long-tail statistics

Nearly half the entire observed vocabulary (47.86%) occurs exactly once. Combined with the extreme concentration of token coverage in a small head (the top 10,000 words, just 0.33% of the vocabulary, cover 82.77% of all tokens), this is a textbook extreme long-tail signature, plausibly amplified by Bengali’s rich inflectional morphology, which fragments a single semantic lemma into many distinct surface word-forms, each individually rare.

Conclusion (Zipf): a single power-law model captures the broad shape of Bengali word frequency but is a measurable oversimplification. The exponent varies by nearly a factor of three across rank regions, and the residual pattern shows structured, non-random deviation. Whether this constitutes a “disproof” of Zipf’s Law depends on how strictly the law is interpreted — as a precise single-exponent claim, the evidence here argues against it; as a general qualitative description of highly skewed frequency distributions, the overall shape is still broadly Zipfian.

4 Heaps’ Law Analysis

Heaps’ Law predicts $V(N) = K \cdot N^\beta$, i.e. vocabulary size grows sub-linearly with corpus size. Vocabulary growth was measured directly via a single memory-safe streaming pass over the corpus, recording 20 logarithmically-spaced checkpoints from 10,000 tokens up to the full 440,930,234-token corpus. The 16 checkpoints at or above 100,000 tokens were used for the regression fit (the smallest 4 were excluded to avoid noisy low- N estimates).

This is a substantially cleaner fit than the Zipf regression: the residuals (not reproduced here for space) are small (roughly 1–7% in log space) and show no strong systematic drift, in contrast to the clearly structured Zipf residuals. Vocabulary grew from 4,416 distinct types at 10,000 tokens

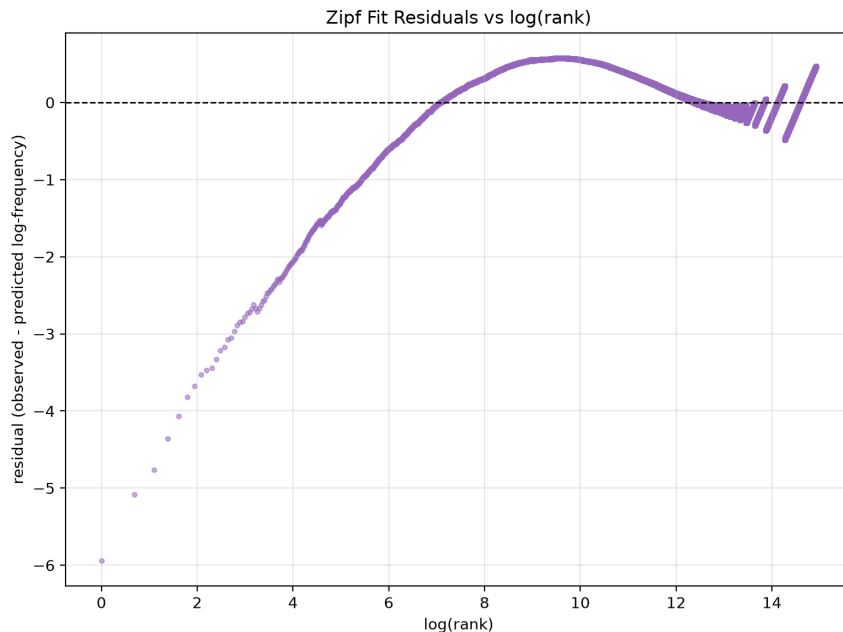


Figure 3: Zipf fit residuals vs. $\log(\text{rank})$. Systematic curvature, not random noise, confirms the single-exponent model is an oversimplification.

to 3,012,823 types at 440.9 million tokens — a $\sim 682\times$ vocabulary increase against a $\sim 44,000\times$ increase in corpus size, consistent with the measured sub-linear exponent.

Critically, the growth curve shows no sign of flattening within the observed range: even at 440.9 million tokens, new distinct word-forms are still appearing at a rate consistent with the fitted power law rather than saturating. Practically, this implies that further data acquisition for Bengali would likely continue to expand vocabulary coverage rather than encountering diminishing returns — a qualitative observation, not a quantitative prediction of how much new vocabulary any specific amount of additional text would contain.

Cross-check against the Zipf results

The classical theoretical relationship between the two laws is $\beta \approx 1/s$. Using the overall Zipf exponent ($s=1.456$) predicts $\beta \approx 0.687$, while the independently measured Heaps exponent is 0.590 — a real, non-trivial gap. Interestingly, the piecewise Zipf exponent for the most frequent 100 words ($s=0.586$) is almost numerically identical to the measured Heaps β (0.590). We note this as a striking empirical coincidence worth reporting, but caution against over-interpreting it: the theoretical $\beta \approx 1/s$ relationship is conventionally derived from tail behaviour of the frequency distribution, not head behaviour, so this correspondence should not be read as validating the theoretical relationship. The more defensible conclusion is the same one Section 3 reaches independently: the single global Zipf exponent does not predict the corpus’s actual vocabulary-growth behaviour well, reinforcing that a single power law is an incomplete description of this corpus.

Statistic	Value
Hapax legomena (freq = 1) as % of vocabulary	47.86%
Vocabulary with frequency ≤ 2	64.78%
Vocabulary with frequency ≤ 5	79.42%
Vocabulary with frequency ≤ 10	86.41%
Token coverage, top 10 words	6.52%
Token coverage, top 100 words	22.40%
Token coverage, top 1,000 words	51.71%
Token coverage, top 10,000 words	82.77%

Table 4: Long-tail and coverage statistics.

K	β	R^2	RMSE (log scale)	Checkpoints used
25.416	0.5903	0.9997	0.0262	16 of 20

Table 5: Fitted Heaps’ Law parameters.

5 Infrastructure & Cost Estimation

This section estimates the cost of extending Bengali support under two neutral architectural archetypes — a **search-oriented system** (crawl/index/serve, representative of Google Search-style infrastructure) and an **LLM-oriented system** (train-or-adapt/fine-tune/serve, representative of Sarvam-style infrastructure) — using a cost model organized into three explicitly separated tiers of evidentiary strength.

SECTION 1 — FACT / ASSIGNMENT INPUT

The assignment’s stated rule — USD 1,000 per 100,000 words — applied directly and without modification across three corpus-size scenarios:

Scenario	Corpus size (words)	Data cost (USD)
SMALL	10,000,000	\$100,000
MEDIUM	100,000,000	\$1,000,000
LARGE	1,000,000,000	\$10,000,000

Table 6: Assignment-based data cost by corpus scenario.

SECTION 2 — SCENARIO ASSUMPTION (architectural cost model)

Beyond the assignment’s data-cost rule, eight cost components (four one-time, four recurring annual) were modelled for each architecture at the MEDIUM (100M word) scenario, each with an explicit justification and an uncertainty range in the supporting assumptions table. **This is a transparent, reproducible scenario model — not Google’s or Sarvam’s actual internal costs.**

At the MEDIUM scenario, totals are **\$1,130,000 initial / \$72,000 recurring per year** for the search-oriented system and **\$1,114,600 initial / \$97,000 recurring per year** for the

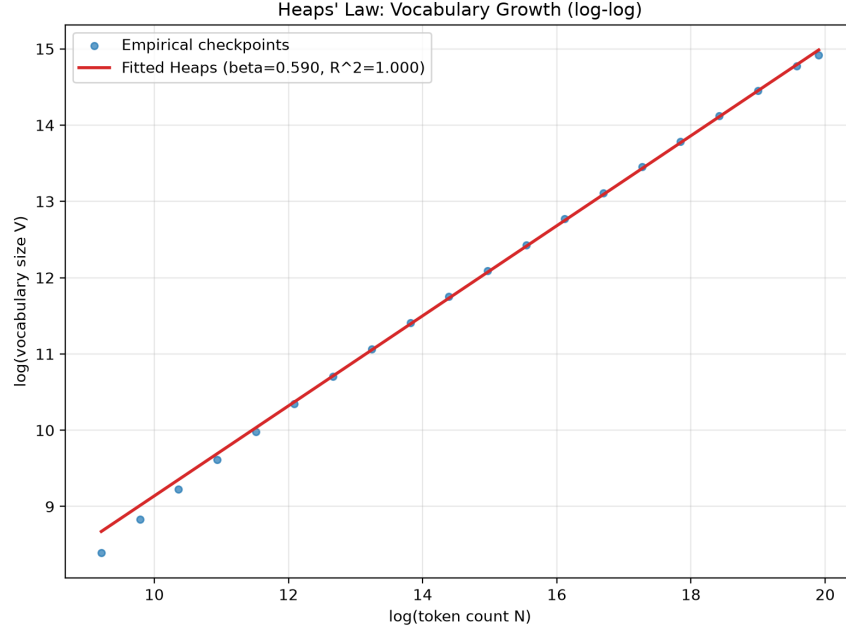


Figure 4: Vocabulary growth (log-log) with fitted Heaps’ Law line. The fit is essentially exact ($R^2 = 0.9997$) across four orders of magnitude of corpus size.

LLM-oriented system. Data acquisition alone accounts for 88.5% (search) and 89.7% (LLM) of initial cost in each architecture — the two architectures differ far more in their *recurring* costs (serving/inference and compute are both notably higher for the LLM-oriented system, reflecting the ongoing cost of running inference versus serving a search index) than in their one-time setup cost.

Three plausible cost components — marketing/user acquisition, legal/regulatory compliance, and customer support operations — were deliberately *excluded* from the model rather than represented with fabricated numbers, because no defensible corpus-size-based scaling rule exists for any of them; they depend on market, jurisdiction, and user-base factors outside this model’s scope.

Sensitivity analysis

SECTION 3 — QUALITATIVE INTERPRETATION (Heaps integration)

Person 3’s fitted Heaps’ Law parameters ($\beta=0.5903$, $K=25.416$) were used to project vocabulary growth at several corpus-size multipliers, purely as narrative context:

This projection is never converted into a cost multiplier anywhere in this project. The cost figures in Sections 1 and 2 above scale strictly with corpus-size scenario (SMALL/MEDIUM/LARGE), independent of any Zipf or Heaps statistic. The qualitative reading is simply that since vocabulary growth has not plateaued (Section 4), continued data acquisition for Bengali would likely keep improving vocabulary coverage — a reason to expect diminishing-but-nonzero returns on future data investment, not a basis for a specific cost figure.

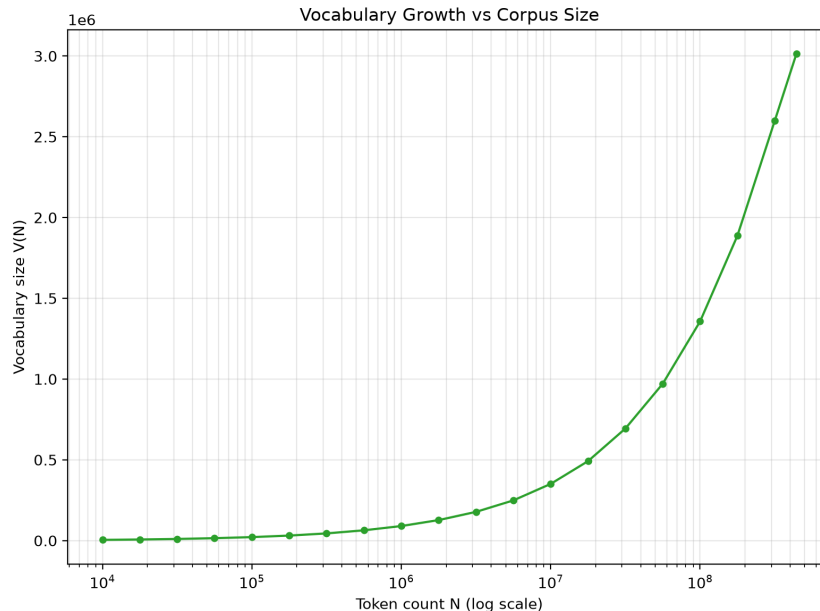


Figure 5: Vocabulary size vs. corpus size (linear scale). Growth has not visibly plateaued even at the largest observed corpus size.

Component	Type	Search-Oriented	LLM-Oriented
Data acquisition	One-time	\$1,000,000	\$1,000,000
Engineering setup	One-time	\$45,000	\$55,000
Indexing / training	One-time	\$50,000	\$17,600
Preprocessing setup	One-time	\$35,000	\$42,000
Compute	Recurring/yr	\$20,000	\$30,000
Maintenance & operations	Recurring/yr	\$15,000	\$18,000
Serving / inference	Recurring/yr	\$25,000	\$40,000
Storage	Recurring/yr	\$12,000	\$9,000

Table 7: Full per-component cost breakdown, MEDIUM scenario.

6 Limitations

- **Sampling.** 10M of ~ 29.6 M rows (33.8%) were processed using sequential prefix sampling from the data stream, not a full-split uniform random sample. Results should generalize to the full split under the assumption that the source corpus is not order-sorted by content type.
- **Tokenization.** A simple whitespace tokenizer was used consistently across preprocessing, Zipf, and Heaps analysis. A linguistically-informed Bengali morphological tokenizer would very likely produce different exponents, vocabulary sizes, and type-token ratios; the reported statistics describe this corpus under this specific tokenization choice, not an intrinsic property of Bengali.
- **Zipf regional boundaries.** The five piecewise rank regions used fixed, pre-specified boundaries rather than automatically detected changeoints; the true deviation structure may not align exactly with these boundaries.

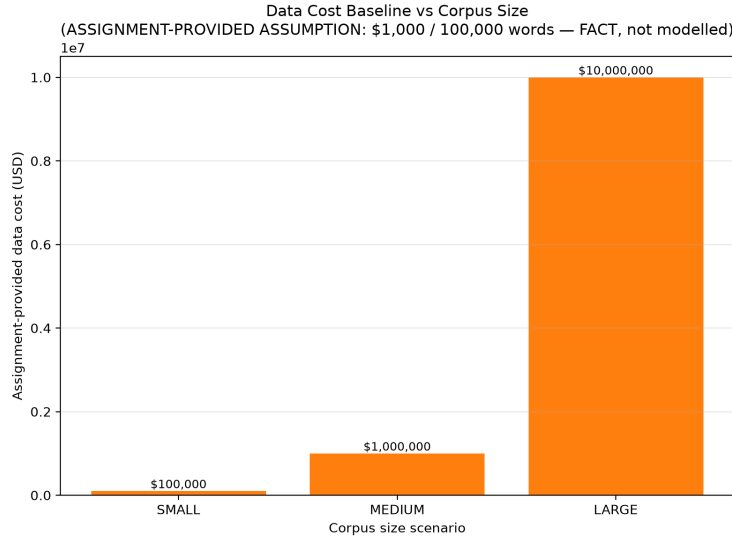


Figure 6: The assignment-provided data cost baseline scales linearly with corpus size by construction; this is a stated fact, not a modelled estimate.

System	Level	Initial (USD)	Recurring/yr (USD)
Search-Oriented	LOW	\$1,091,250	\$49,750
Search-Oriented	BASE	\$1,130,000	\$72,000
Search-Oriented	HIGH	\$1,168,750	\$94,250
LLM-Oriented	LOW	\$1,079,230	\$66,350
LLM-Oriented	BASE	\$1,114,600	\$97,000
LLM-Oriented	HIGH	\$1,149,970	\$127,650

Table 8: LOW/BASE/HIGH sensitivity bounds, MEDIUM scenario.

- **Cost model scope.** The Section 5 cost figures are a transparent, reproducible architectural scenario model built on labelled, documented assumptions with stated uncertainty ranges (typically 20–35%) — they are explicitly not Google’s, Sarvam’s, or any other company’s actual internal costs.

7 Conclusion

Across a 10-million-row, ~441-million-token sample of Bengali web text, a single Zipfian exponent ($s=1.456$) fits the overall rank-frequency distribution reasonably well ($R^2=0.976$) but conceals substantial and systematic regional variation (s ranging 0.586–1.630 across rank bands), indicating that Bengali word frequency is better described as approximately Zipfian with structured deviations than as following one clean power law. Vocabulary growth, by contrast, is very well described by Heaps’ Law ($\beta=0.590$, $R^2=0.9997$) with no plateau observed at this scale, implying continued data acquisition would likely keep expanding Bengali vocabulary coverage. On the infrastructure side, applying the assignment’s linear data-cost assumption alongside an independently documented architectural cost model shows that, at a 100-million-word scale, data

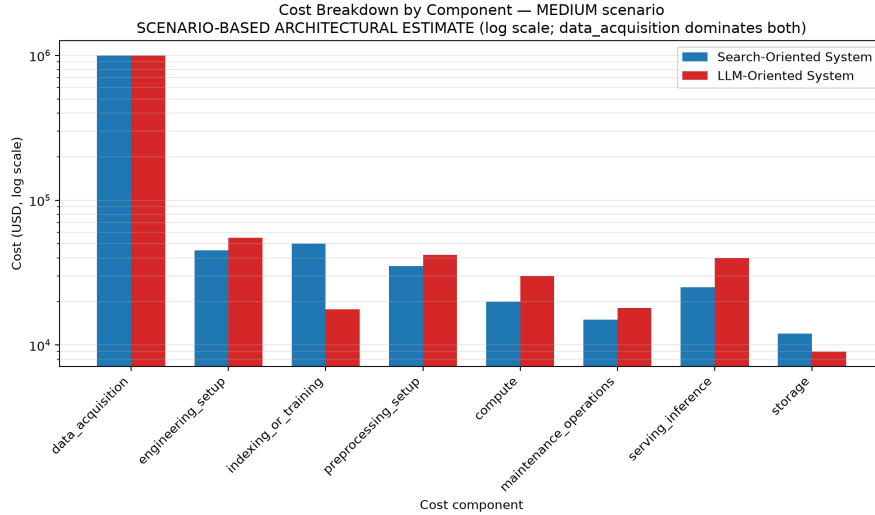


Figure 7: Full per-component cost breakdown (log scale), MEDIUM scenario. Data acquisition dominates both architectures by over an order of magnitude.

Corpus size multiplier	Projected vocabulary	Growth factor
2×	4,536,029	1.51×
5×	7,790,858	2.59×
10×	11,729,715	3.89×
100×	45,666,872	15.16×

Table 9: Heaps’-Law-projected vocabulary growth at increasing corpus-size multipliers.

acquisition itself dominates one-time investment for both a search-oriented and an LLM-oriented system (each over 88% of initial cost), while the two architectures diverge more meaningfully in ongoing operational cost than in upfront investment.

References

- AI4Bharat. *IndicCorpV2* [dataset]. Hugging Face. <https://huggingface.co/datasets/ai4bharat/IndicCorpV2>

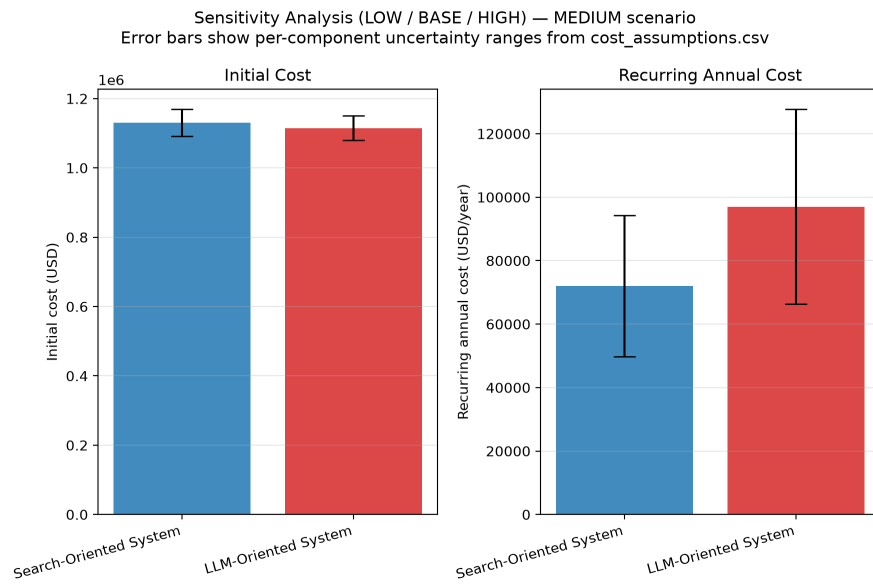


Figure 8: LOW/BASE/HIGH sensitivity ranges. Initial cost is tightly bounded (dominated by the fixed data-acquisition figure); recurring cost carries proportionally wider uncertainty, especially for the LLM-oriented system.